Chapter 5
Differentiation

Note 1

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5.1 : The Derivative

In this section, we will present some of the elementary properties of the derivative. We begin with the definition of the derivative of a function.

Definition 5.1.1 : A function f is said to be ***differentiable*** at a point $a \in \mathbb{R}$ if and only if f is defined on some open interval I containing a and

$$f'(a) := \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{1}$$

exists. In this case $f'(a)$ is called the ***derivative*** of f at a.

Note : The assumption that f be defined on an open interval containing a is made so that the quotients in (1) are defined for all $h \neq 0$ sufficiently small.

Geometric Interpretation of (1) :

- Suppose that f is differentiable at a. A ***secant line*** of the graph $y = f(x)$ is a line passing through at least two points on the graph, and a ***chord*** is a line segment which runs from one point on the graph to another.
- Let $x = a + h$, and observe that the ***slope*** of the chord passing through the points $(x, f(x))$ and $(a, f(a))$ is given by $\dfrac{(f(x) - f(a))}{(x - a)}$.
- Now, since $x = a + h$, (1) becomes

$$f'(a) = \frac{f(x) - f(a)}{x - a} \ .$$

- Hence, as $x \to a$ the slopes of the chords through $(x, f(x))$ and $(a, f(a))$ approximate the slope of the tangent line of $y = f(x)$ at $x = a$ (see **Figure 6.1.1**), and in the limit, the slope of the tangent line to $y = f(x)$ at $x = a$ is precisely $f'(a)$.

- Thus, we shall say that the graph of $y = f(x)$ has a unique tangent line at a point $(a, f(a))$ if and only if $f'(a)$ exists.

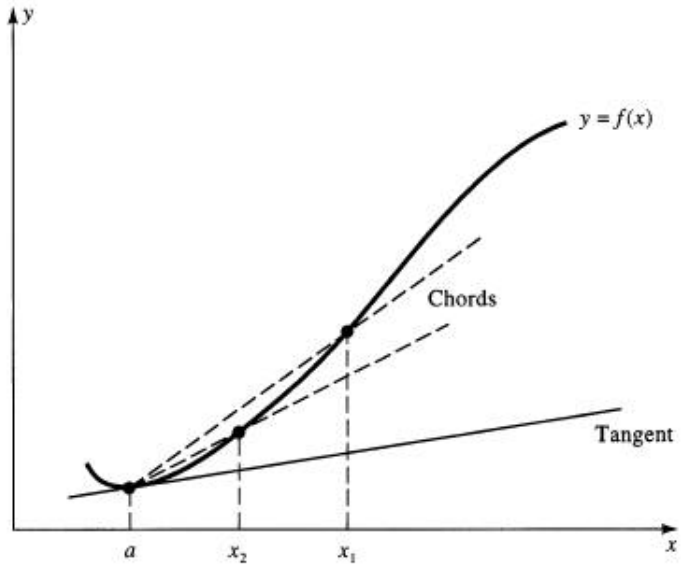


Figure 5.1.1

- When $y = f(x)$, we shall also use the notation dy/dx or y' for f'. Higher-order derivatives are defined recursively; i.e., if $n \in \mathbb{Z}^+$, then $f^{(n+1)}(a) := \left(f^{(n)}\right)'(a)$, provided these derivatives exist.

- Here are two characterizations of differentiability which we shall use to study derivatives. The first one, which characterizes the derivative in terms of the ***chord function***

$$F(x) := \frac{f(x) - f(a)}{x - a}, \quad x \neq a \tag{2}$$

will be used to establish the Chain Rule in Section 5.2.

Theorem 5.1.1 : A function f is differentiable at some point $a \in \mathbb{R}$ if and only if there exists an open interval I and a function $F : I \to \mathbb{R}$ such that $a \in I$, f is defined on I, F is continuous at a, and

$$f(x) = F(x)(x - a) + f(a) \tag{3}$$

holds for all $x \in I$, in which case $F(a) = f'(a)$.

Proof of Theorem 5.1.1.

"$\Longrightarrow$":

- Note that for $x \in I \backslash \{a\}$, (2) and (3) are equivalent.
- Suppose that f is differentiable at a. Then f is defined on some open interval I containing a, and the limit in (1) exists.
- Define F on I by (2) if $x \neq a$, and by $F(a) := f'(a)$. Then (3) holds for all $x \in I$, and F is continuous at a by (2) since $f'(a)$ exists.

"$\Longleftarrow$" :

- If (3) holds, then (2) holds for all $x \in I, x \neq a$.
- Taking the limit of (2) as $x \to a$, since F is continuous at a, we conclude that $F(a) = f'(a)$.

□

The second characterization of differentiability, in terms of ***linear approximations*** (i.e., how well $f(a+h) - f(a)$ can be approximated by a straight line through the origin).

Theorem 5.1.2 : A function f is differentiable at a if and only if there is a function T of the form $T(x) := mx$ such that

$$\lim_{h \to 0} \frac{f(a+h) - f(a) - T(h)}{h} = 0 \ . \tag{4}$$

Proof of Theorem 5.1.2.

Suppose that f is differentiable, and set $m := f'(a)$. Then by (1),

$$\frac{f(a+h) - f(a) - T(h)}{h} = \frac{f(a+h) - f(a)}{h} - f'(a) \longrightarrow 0 \quad \text{as} \ \ h \to 0 \ .$$

Conversely, if (4) holds for $T(x) := mx$ and $h \neq 0$, then

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= m + \frac{f(a+h) - f(a) - mh}{h} \\ &= m + \frac{f(a+h) - f(a) - T(h)}{h} \ . \end{aligned}$$

By (4), the limit of this last expression is m. It follows that $(f(a+h)- f(a))/h \to m$, as $h \to 0$; that is, that $f'(a)$ exists and equals m.

□

Our first application of **Theorem 5.1.2** answers the question: Are differentiability and continuity related ?

Theorem 5.1.3 : If f is differentiable at a, then f is continuous at a.

Proof of Theorem 5.1.3.

- Suppose that f is differentiable at a.
- By **Theorem 5.1.1**, there is an open interval I and a function F, continuous at a, such that $f(x) = f(a) + F(x)\ (x - a)$ for all $x \in I$.
- Taking the limit of this last expression as $x \to a$, we see that

$$\lim_{x\to a} f(x) = f(a) + F(a) \cdot 0 = f(a)\ .$$

- In particular, $f(x) \to f(a)$ as $x \to a$; that is, f is continuous at a.

□

Thus any function which fails to be continuous at a cannot be differentiable at a. The following example shows that the converse of **Theorem 6.1.3** is false.

Example 1 Show that $f(x) = |x|$ is continuous at 0 but not differentiable there.

Proof.

- Since $x \to 0$ implies $|x| \to 0$, f is continuous at 0.
- On the other hand, since $|h| = h$ when $h > 0$ and $|h| = -h$ when $h < 0$, we have

$$\lim_{h\to 0^+} \frac{f(h) - f(0)}{h} = 1 \text{ and } \lim_{h\to 0^-} \frac{f(h) - f(0)}{h} = -1.$$

- Since a limit exists if and only if its one-sided limits exist and are equal (**Theorem 4.2.1**), it follows that the limit in (1) does not exist when $a = 0$ and $f(x) = |x|$. Therefore, f is not differentiable at 0.

□

By **Definition 5.1.1**, if f is differentiable at a, then f must be defined on an open interval containing a (i.e., on both sides of a). As with the theory of limits, it is convenient to define "one-sided" derivatives to deal with functions whose domains are closed intervals (see **Example 3** below). Here is a brief discussion of what it means for a function to be differentiable on an interval (as opposed to being differentiable at every point in an interval).

Definition 5.1.2 :

(i). A function $f : I \to \mathbb{R}$ is said to be ***differentiable*** on I if and only if

$$f'_I(a) := \lim_{\substack{x \to a \\ x \in I}} \frac{f(x) - f(a)}{x - a}$$

exists and is finite for every $a \in I$.

(ii). f is ***continuously differentiable*** on I if and only if f'_I exists and is continuous on I.

Note : When a is not an endpoint of I, $f'_I(a)$ is the same as $f'(a)$. Because of this, we usually drop the subscript on f'_I. In particular, if f is differentiable on $[a, b]$, then

$$f'(a) := \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h} \text{ and } f'(b) := \lim_{h \to 0^-} \frac{f(b+h) - f(b)}{h} .$$

The following example shows that **Definition 5.1.2** enlarges the collection of differentiable functions.

Example 2 The function $f(x) = x^{3/2}$ is differentiable on $[0, \infty)$ and $f'(x) = \frac{3}{2}\sqrt{x}$ for all $x \in [0, \infty)$.

Proof. By the Power Rule, $f'(x) = \frac{3}{2}\sqrt{x}$ for all $x \in (0, \infty)$ and by definition,

$$f'(0) = \lim_{h \to 0^+} \frac{h^{3/2} - 0}{h} = \lim_{h \to 0^+} \sqrt{h} = 0 .$$

□

Here is a notation widely used in conjunction with **Definition 5.1.2**. Let I be an interval. For each $n \in \mathbb{N}$, define the collection of functions $\mathcal{C}^n(I)$ by

$$\mathcal{C}^n(I) \coloneqq \left\{ f : f : I \to \mathbb{R} \text{ and } f^{(n)} \text{ exists and is continuous on } I \right\}.$$

- $\mathcal{C}^1(I)$ is the collection continuously differentiable functions on I.
- The following example shows that not every function which is differentiable on $\mathbb{R}$ belongs to $\mathcal{C}^1(\mathbb{R})$.

Example 3 The function

$$f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0, \\ 0 & x = 0. \end{cases}$$

is differentiable on $\mathbb{R}$ but not continuously differentiable on any interval which contains the origin.

Proof. By definition,

$$f'(0) = \lim_{h \to 0} h \sin\left(\frac{1}{h}\right) = 0 \quad \text{and} \quad f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) \quad \text{for} \quad x \neq 0 \, .$$

Thus f is differentiable on $\mathbb{R}$ but $\lim_{x \to 0} f'(x)$ does not exist. In particular, f' is not continuous on any interval which contains the origin. □

It is important to notice that a function which is differentiable on two sets is not necessarily differentiable on their union.

Remark : $f(x) = |x|$ is differentiable on $[0, 1]$ and on $[-1, 0]$ but not on $[-1, 1]$.

Proof. Since $f(x) = x$ when $x > 0$ and $f(x) = -x$ when $x < 0$, it is clear that f is differentiable on $[-1, 0) \cup (0, 1]$ (with $f'(x) = 1$ for $x > 0$ and $f'(x) = -1$ for $x < 0$) . By Example 1, f is not differentiable at $x = 0$. However,

$$f'_{[0,1]}(0) = \lim_{h \to 0^+} \frac{|h|}{h} = 1 \quad \text{and} \quad f'_{[-1,0]}(0) = \lim_{h \to 0^-} \frac{|h|}{h} = -1.$$

Therefore, f is differentiable on $[0, 1]$ and on $[-1, 0]$. □

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EXERCISES

(1). For each of the following real functions, use **Definition 5.1.1** directly to prove that $f'(a)$ exists.

(a) $f(x) = x^2 + x, a \in \mathbb{R}$

(b) $f(x) = \sqrt{x}, a > 0$

(c) $f(x) = 1/x, a \neq 0$

(2). Let I be an open interval which contains 0 and $f : I \to \mathbb{R}$. If there exists an $\alpha > 1$ such that $|f(x)| \leq |x|^\alpha$ for all $x \in I$, prove that f is differentiable at 0. What happens when $\alpha = 1$?

(3). (a) Find all points (a, b) on the curve C, given by $y = x + \sin x$, so that the tangent lines to C at (a, b) are parallel to the line $y = x + 15$.

(b) Find all points (a, b) on the curve C, given by $y = 3x^2 + 2$, so that the tangent lines to C at (a, b) pass through the point (-1,-7).

(4). Suppose that $f : (0, \infty) \to \mathbb{R}$ satisfies $f(x) - f(y) = f(x/y)$ for all $x, y \in (0, \infty)$ and $f(1) = 0$.

(a) Prove that f is continuous on $(0, \infty)$ if and only if f is continuous at 1.

(b) Prove that f is differentiable on $(0, \infty)$ if and only if f is differentiable at 1.

(c) Prove that if f is differentiable at 1, then $f'(x) = f'(1)/x$ for all $x \in (0, \infty)$.

[Note: If $f'(1) = 1$, then $f(x) = \log x$.]

(5). Let I be an open interval, $f : I \to \mathbb{R}$, and $c \in I$. The function f is said to have a *local maximum* at c if and only if there is a $\delta > 0$ such that $f(c) \geq f(x)$ holds for all $|x - c| < \delta$.

(a) If f has a local maximum at c, prove that

$$\frac{f(c+u) - f(c)}{u} \leq 0 \quad \text{and} \quad \frac{f(c+t) - f(c)}{t} \geq 0$$

for $u > 0$ and $t < 0$ sufficiently small.

(b) If f is differentiable at c and has a local maximum at c, prove that $f'(c) = 0$.

(c) Make and prove analogous statements for local minima.

(d) Show by example that the converses of the statements in parts (b) and (c) are false. Namely, find an f such that $f'(0) = 0$ but f has neither a local maximum nor a local minimum at 0.

(6). Suppose that $I = (-a, a)$ for some $a > 0$. A function $f : I \to \mathbb{R}$ is said to be *even* if and only if $f(-x) = f(x)$ for all $x \in I$, and said to be *odd* if and only if $f(-x) = -f(x)$ for all $x \in I$.

(a) Prove that if f is odd and differentiable on I, then f' is even on I.

(b) Prove that if f is even and differentiable on I, then f' is odd on I.
